Ru Peng (彭儒)

CS Ph.D., DI-Lab, Zhejiang University
Personal Webpage: Homepage ◇ Google Scholar ◇ Github
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Research Interests: LLMs, Evaluation, Multimodal, Machine Translation.

EDUCATION

Zhejiang University *Sep 2022 - Present*
Ph.D. of Computer Science Supervisor: Prof. Junbo Zhao

Guangdong University of Technology *Sep 2017 - Jul 2020*
M.S. of Communication Engineering Supervisors: Prof. Yi Fang and Prof. Tianyong Hao

Liaoning Technical University *Sep 2013 - Jul 2017*
B.E. of Communication Engineering GPA: 81/100, rank 16/122

WORK EXPERIENCE

Institute of Computer Innovation, Zhejiang University *Sep 2021 - Sep 2022*
Research Intern on Multimodal Machine Translation and Automated Model Evaluation
Mentored by Prof. Junbo Zhao and Dr. Peng Lu

ATTL Laboratoroty, NICT, Japan *Dec 2020 - Jun 2021*
Research Intern (Remote) on Textual, Multimodal and Sign Language Translation
Mentored by Prof. Kehai Chen

Text Analytics and Mining Lab, South China Normal University *Jul 2018 - Jul 2020*
Cooperative Student on Textual Machine Translation
Mentored by Prof. Tianyong Hao

PUBLICATIONS

Machine Translation

[1] Deps-SAN: Neural Machine Translation with Dependency-Scaled Self-Attention Network.
Ru Peng, Nankai Lin, Yi Fang, Shengyi Jiang, Tianyong Hao, Boyu Chen, Junbo Zhao.
The 29th International Conference on Neural Information Processing (**ICONIP**), 2022. **(Oral)**

[2] Syntax-Aware Attentional Neural Machine Translation Directed by Syntactic Dependency Degree.
Ru Peng, Tianyong Hao, Yi Fang.
Neural Computing and Applications (**NCA**), 2020. **(IF: 5.102)**

[3] Neural Machine Translation with Attention Based on a New Syntactic Branch Distance.
Ru Peng, Zhitao Chen, Tianyong Hao, Yi Fang.
The 15th China Conference on Machine Translation (**CCMT**), 2019. **(Best Paper Candidates)**

Multimodal

[4] Better Sign Language Translation with Monolingual Data.
Ru Peng, Yawen Zeng, Junbo Zhao.
arXiv:2304.10844, 2023.

[5] Distill The Image to Nowhere: Inversion Knowledge Distillation for Multimodal Machine Translation.
Ru Peng, Yawen Zeng, Junbo Zhao.
The 2022 Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2022. **(Oral, top 4.1%)**

[6] HybridVocab: Towards Multi-Modal Machine Translation via Multi-Aspect Alignment.
Ru Peng, Yawen Zeng, Junbo Zhao.
International Conference on Multimedia Retrieval (**ICMR**), 2022. **(Oral)**

Evaluation

[7] CAME: Contrastive Automated Model Evaluation.
Ru Peng, Qiuyang Duan, Haobo Wang, Jiachen Ma, Yanbo Jiang, Yongjun Tu, Xiu Jiang, Junbo Zhao.
IEEE/CVF International Conference on Computer Vision (**ICCV**), 2023.

[8] One Chinese patent application work with OPPO in submitted, 3rd author.

SKILLS

Programming Languages: Java, Python, C, C#, Matlab, Shell, Perl
Deep Learning Frameworks: Pytorch, Tensorflow, Theano, Keras
Machine Learning Libraries: NumPy, SciPy, Sklearn, Matplotlib, Pandas
AI Experiences: Theories, Toolkits, Codebases, Experiments for NLP/CV/Multimodal/ML fields

AWARDS

Merit Student, Zhejiang University, 2022
Academic Scholarship, GDUT, 2017 – 2019
Outstanding Student Scholarship, LNTU, 2016

PROFESSIONAL SERVICE

Conference Reviewer: NeurIPS 2022 - 2023, ICML 2023, ICCV 2023, ICLR 2024